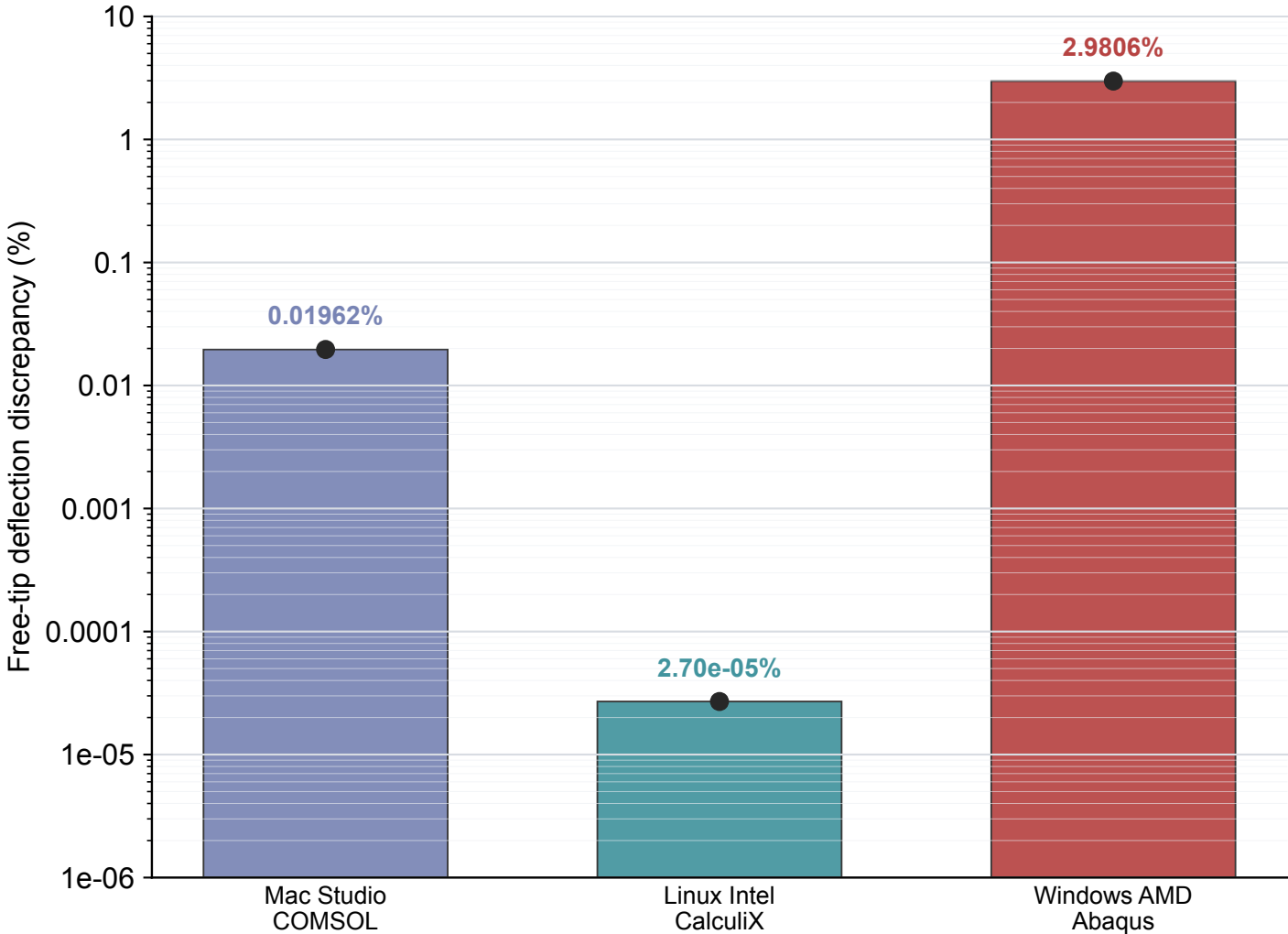


Structured Hex8/C3D8 cantilever

a Free-tip deflection discrepancy between MATLAB solve and FE reference
Cross-platform solve-level validation



Source data: macOS+COMSOL legacy report table; Linux+CalculiX validation report; Windows+Abaqus free-tip summary CSV. Sparse-pattern asset recorded in source_manifest.json.

b

Metric

$100 \times \frac{||U_{z, \text{MATLAB}}| - |U_{z, \text{FE}}||}{|U_{z, \text{FE}}|}$
probe: free_tip_center
quantity: vertical displacement Uz
scale: percent, log axis

Case metadata

nodes: 325
elements: 192
DOFs: 975
nnz(K): 56277
E = 1, nu = 0.3
L = 1, W = 0.2, T = 0.1

Interpretation

Abaqus shows a 2.98% Hex8 discrepancy; COMSOL and CalculiX stay below 0.02%.
Solver-validation case: cantilever_hex8_medium.